

Math 104 – Real Analysis

Homework 4 — my solutions

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Homework 4

Deadline: September 27

All sequences are real-valued.

Problem 1.

Suppose a sequence (s_n) satisfies

$$|s_{n+1} - s_n| \leq \frac{1}{n 2^n} \quad (n \in \mathbb{N}).$$

Prove that (s_n) is Cauchy and therefore has a limit $L \in \mathbb{R}$. Then prove that, for every $n \in \mathbb{N}$,

$$|s_n - L| \leq \frac{1}{n 2^{n-1}}.$$

Use finite geometric sums in your Cauchy argument, not a convergence test for infinite series.

Problem 2.

Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and suppose $\alpha = \sup A$ does not belong to A . Construct a *strictly increasing* sequence (a_n) of points of A such that

$$0 < \alpha - a_n < 2^{-n} \quad (n \in \mathbb{N}).$$

Justify both the construction and the conclusion $a_n \rightarrow \alpha$.

Problem 3.

Define

$$x_1 = 4, \quad x_{n+1} = \frac{3x_n + 4}{x_n + 3} \quad (n \geq 1).$$

Prove that the recurrence is well defined, that $2 < x_n \leq 4$ for every n , and that (x_n) is strictly decreasing. Deduce that (x_n) converges and determine its limit. Establish convergence before taking limits in the recurrence.

Problem 4.

Let

$$a_n = 2 + (-1)^n + \frac{1 + (-1)^n}{n} \quad (n \in \mathbb{N}).$$

Exhibit a constant subsequence and a convergent, nonconstant subsequence with a different limit. Use these subsequences to decide whether (a_n) converges, and justify your conclusion.

Problem 5.

Let $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ be strictly increasing, and suppose that $s_{\sigma(k)} \rightarrow L \in \mathbb{R}$. Construct a strictly increasing map $\tau : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$|s_{\sigma(\tau(j))} - L| < 2^{-j} \quad (j \in \mathbb{N}).$$

Prove explicitly that $\sigma \circ \tau$ is strictly increasing. Conclude that $(s_{\sigma(\tau(j))})$ is a subsequence of the original sequence (s_n) and converges to L .

Solutions

Problem 1

Solution. We do this using triangle inequality. For our scratch work, Let's bound

$$|s_m - s_n| = |(s_m - s_{m-1}) + (s_{m-1} - s_{m-2}) + \cdots + (s_{n+1} - s_n)| \leq \sum_{i=n}^{m-1} |s_{i+1} - s_i| \leq \sum_{i=n}^{m-1} \frac{1}{i2^i}$$

Since $n \geq 1$, notice that $\frac{1}{i2^i} \leq \frac{1}{2^i}$ and so we'll bound again,

$$\sum_{i=n}^{m-1} \frac{1}{i2^i} \leq \sum_{i=n}^{m-1} \frac{1}{2^i} = \frac{1}{2^{n-1}} - \frac{1}{2^{m-1}} < \frac{1}{2^{n-1}} < \epsilon$$

So if we rearrange, we can write $N = \max \{1, \lceil \log_2 \frac{1}{\epsilon} \rceil + 2\}$, we have the Cauchy condition, namely for an $\epsilon > 0$,

$$\forall n, m \geq N \implies |s_m - s_n| < \epsilon$$

and thus it has a limit $L \in \mathbb{R}$.

Notice if $\lim s_n = L$, then for each fixed n , $s_m \rightarrow L$, so

$$|s_n - L| = \lim_{m \rightarrow \infty} |s_n - s_m|.$$

To do this, we'll use the same triangle inequality idea

$$|s_n - s_m| \leq \sum_{i=n}^{m-1} |s_{i+1} - s_i| \leq \sum_{i=n}^{m-1} \frac{1}{i2^i} \leq \frac{1}{n} \sum_{i=n}^{m-1} \frac{1}{2^i}$$

And this is really convenient as this sum is equal to

$$|s_m - s_n| \leq \frac{1}{n} \left(\frac{1}{2^{n-1}} - \frac{1}{2^{m-1}} \right).$$

Taking $m \rightarrow \infty$, we get

$$|s_n - L| \leq \frac{1}{n} \left(\frac{1}{2^{n-1}} \right) = \frac{1}{n2^{n-1}},$$

as desired. □

Problem 2

Solution. To do this Let's first rearrange the condition to get $\alpha > a_n > \alpha - 2^{-n}$

We can construct it as follows. For a_1 we have $\alpha > a_1 > \alpha - \frac{1}{2}$. Then, for each $i \geq 2$, we can find

$$\alpha > a_i > \max\{a_{i-1}, \alpha - 2^{-i}\}.$$

This is the construction, we claim that it's valid along the following lines. Because $a_{i-1} \in A$ and $\alpha \notin A$, we have $a_{i-1} < \alpha$, and also $\alpha - 2^{-i} < \alpha$. Thus

$$\max\{a_{i-1}, \alpha - 2^{-i}\} < \alpha.$$

By the supremum property,

$$\exists a_i \in A \text{ such that } \alpha > a_i > \max\{a_{i-1}, \alpha - 2^{-i}\},$$

and that justifies our construction. In particular, $a_i > a_{i-1}$, so the sequence is strictly increasing. Because $\alpha - 2^{-n} < a_n < \alpha$ and $\alpha - 2^{-n} \rightarrow \alpha$, by the squeeze theorem, $a_n \rightarrow \alpha$. $\square$

Problem 3

Solution. For this, let's first show $x_n > 2$ by induction.

Our base case is $x_1 = 4 > 2$.

For the inductive step, suppose $x_k > 2$ for an arbitrary $k \in \mathbb{Z}^+$. Since $x_k + 3 > 0$, the recurrence is well defined, and

$$x_{k+1} - 2 = \frac{3x_k + 4}{x_k + 3} - 2 = \frac{x_k - 2}{x_k + 3} > 0.$$

Thus $x_{k+1} > 2$, so $x_n > 2$ for every n .

Let's show $x_{k+1} < x_k$.

$$x_{k+1} - x_k = \frac{3x_k + 4}{x_k + 3} - x_k = \frac{4 - x_k^2}{x_k + 3} < 0,$$

since $x_k > 2$, so $4 - x_k^2 < 0$, while $x_k + 3 > 0$. Thus (x_k) is strictly decreasing. Since $x_1 = 4$, we have $2 < x_n \leq 4$ for every n .

Because it is decreasing and bounded below, $\lim_{k \rightarrow \infty} x_k = L$ exists.

Notice similarly $\lim_{k \rightarrow \infty} x_{k+1} = L$, so we write

$$L = \lim_{k \rightarrow \infty} x_{k+1} = \lim_{k \rightarrow \infty} \frac{3x_k + 4}{x_k + 3}.$$

Because $x_k > 2$ for every k , we have $L \geq 2$, so $L + 3 \neq 0$. Therefore

$$L = \frac{\lim(3x_k + 4)}{\lim(x_k + 3)} = \frac{3L + 4}{L + 3}.$$

And solving this quadratic for L gets you

$$L^2 + 3L = 3L + 4 \rightarrow L = \pm 2.$$

Since $L \geq 2$, we get $L = 2$. $\square$

Problem 4

Solution. If you examine the even elements, you see that

$$\lim(a_{2k}) = \lim(2 + (-1)^{2k}) + \lim \frac{1 + (-1)^{2k}}{2k} = 3 + 0 = 3$$

However, if you examine the odd elements, you get

$$\lim(a_{2k+1}) = \lim(2 + (-1)^{2k+1}) + \lim \frac{1 + (-1)^{2k+1}}{2k+1} = 1 + 0 = 1$$

Notice the odd sequence is constant, because $2 + (-1)^{2k+1} = 1$ and $1 + (-1)^{2k+1} = 0$, but the even sequence is convergent but non constant, because each term $a_{2k} = 3 + \frac{1+(-1)^{2k}}{2k} = 3 + \frac{1}{k}$. If (a_n) converged, then every subsequence of (a_n) would converge to the same limit. Since the odd subsequence converges to 1 and the even subsequence converges to 3, (a_n) does not converge. $\square$

Problem 5

Solution. We want a sequence such that

$$|s_{\sigma(\tau(j))} - L| < 2^{-j}$$

We can arrange this to

$$L - 2^{-j} < s_{\sigma(\tau(j))} < L + 2^{-j}.$$

Since $s_{\sigma(k)} \rightarrow L$, we can make a sequence $N_1, N_2, \dots$ where, for a specific j , we can set $\epsilon = 2^{-j}$ and have

$$\exists N_j \text{ such that } k \geq N_j \implies |s_{\sigma(k)} - L| < \epsilon = 2^{-j}$$

for each j in $\mathbb{Z}^+$. To continue, set $\tau(1) = N_1$, and for $i \geq 2$ recursively construct

$$\tau(i) = \max\{N_i, \tau(i-1) + 1\}.$$

The N_i ensures that $|s_{\sigma(\tau(i))} - L| < 2^{-i}$ and the other term ensures that $\tau(i) > \tau(i-1)$, so τ is strictly increasing. We have that σ is strictly increasing and since our constructed τ is strictly increasing, we have that

$$j > i \implies \tau(j) > \tau(i) \implies \sigma(\tau(j)) > \sigma(\tau(i)),$$

so $\sigma \circ \tau$ is strictly increasing. Thus, $(s_{\sigma(\tau(j))})$ is a subsequence of the original sequence (s_n) and, since

$$0 \leq |s_{\sigma(\tau(j))} - L| < 2^{-j} \rightarrow 0,$$

it converges to L . $\square$